

Internet for X

Fabián E. Bustamante
Northwestern

Contributors: Radhika Mittal, Mlab, Keith Winstein, Y. Richard Yang,
Walter Willinger

Common threads ...

- New abstraction for a new Internet
- Managing resources in good and bad times
- Gaining confidence

New
abstraction

Managing
resources

Gaining
confidence

App-aware resource management (Radhika)

- Higher demand and stricter requirement on last-mile from apps (e.g., AR/VR)
- Last mile infrastructure (wired or wireless) has not kept up with it
- Net resource mgmt is well studied, but current mechs lack app awareness
 - A philosophical barrier that requires the network to be general and neutral
 - A deployment barrier that makes it difficult to apply this
- To avoid this
 - Decouple decision-making from enforcement (policy/mechanism) - Parameterized abstractions that let users and apps decide/control their own traffic
 - Limit deployment to the last mile - easier to do and where the bottleneck is

New
abstraction

Managing
resources

App-defined network resource allocation (Richard)

- Network resource allocation in the Internet is largely based on TCP CC, based on universal fairness
- But in many settings like CERN's LHCONE*, it is more important to allocate resources based on app need
- One solution - A flipped control architecture where
 - Network provide resource control requirements
 - App control plane realizes global app-driven resource allocation
- More general
 - What abstraction should the network infrastructure provide? The app control plane should have visibility into how app-level resource usage is mapped into infrastructure resource usage
 - How should app-defined resource allocation interact with network infrastructure adaptation?

*Large Hadron Collider Open Network Environment, an L3VPN network connecting 600 distributed storage systems, distributed across 170 data centers in 40 countries

New
abstraction

Managing
resources

Designing network infrastructure for computation (Keith)

- Providers' model is one of renting (virtual) hw (or paying for effort)
- Providers have little visibility into applications dataflows
 - Tasks spend time waiting for data, but dataflow is revealed only after placement, while executing
- Computation centric networking
 - The network is there to provide answers to computation
 - Visibility into the dataflow
 - An objective, common notion of correctness
 - Separation between i/o and compute
- It could allow
 - Ultra-high-density multi tenancy, accommodate heterogeneous infrastructure, massive burst parallelism, verifiable computation
- Changing to 'paying for results'

New
abstraction

Managing
resources

Internet survivability (Walter, Fabián)

- A critical cyber-physical infrastructure not designed as such
- Assumption - The so-far robust current architecture will hold
- But,
 - As Internet becomes more critical, higher risks/impact of disruption
 - And the frequency and scale of many disrupting events increase
- A new research agenda centered on Internet resilience
 - An understanding of architectural principles on which it should be based
 - E.g., Revisiting the network control system architecture based on insights from multi-layers sensorimotor control systems
 - A reassessment of threats (endogenous and exogenous)
 - Must be done in concert with closely interconnected critical systems (e.g., power grid)

Managing
resources

Gaining
confidence

Public decision flows (Keith)

- Most/all reinforcement algorithms do not get deployed
- Part of the problem
 - The Internet is a uniquely challenging setting, and we don't know how to simulate it
- RL needs to happen in simulation
 - Too slow to train an algorithm in the real network
- Idea - proceed in stages with
 1. Simulators and emulators as hypothesis generators and first stage at selection
 2. Real-world testing of a few algorithms with a small frac of data
 - a. No datasets or dataflows, but decision flows - *A sequence of opportunities to make decisions that affect representative network traffic in a randomized fashion*

E.g., For ABR, 0.01% of streaming video traffic of Netflix, but under the condition that it must experience 95% of QoE of untouched traffic

New
abstraction

Gaining
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Verifiable monitoring data (MLab)

- To evaluate the quality of Internet connectivity we need publicly available, verifiable data and info collected over time
- Old problem but the last pandemic brought renewed interest
- But
 - There's nothing like it (typically is done by private companies)
 - There's no single tool we all agree on
 - Way to do it today is to drop a device that needs to be paid for, maintained, troubleshooted, ...
 - Without a company, there is no financial incentive and no clear path to sustainability
 - Debugging home networking is becoming even harder with more home devices
- MLab can help with this

Gaining
confidence

Common threads

- New abstraction for a new Internet
 - On the last mile
 - On domain-specific networks
 - For cloud/infrastructure providers
- Managing resources in good and bad times (need new abstractions)
 - Under the stress of natural or man-made disasters
 - When too many demanding applications require it
 - When needs deviate from universal fairness
- Gaining confidence
 - Through publicly available, verifiable monitoring data
 - Via testing in the wild
 - *What we can measure is often not what we want to measure*